



MATH240: Undetermined Coefficients

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Particular solution — constant forcing

1. Find a particular solution y_p for $y'' + 6y' + 1y = 2$.

$$y'' + 6y' + 1y = 2$$

Answer: _____

2. A spring with damping satisfies $y'' + 2y' + 4y = 8$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 8$$

Answer: _____

3. Find the steady-state (particular) solution of $y'' + 4y' + 7y = 20$.

$$y'' + 4y' + 7y = 20$$

Answer: _____

4. Find a particular solution y_p for $y'' + 3y' + 4y = 17$.

$$y'' + 3y' + 4y = 17$$

Answer: _____

5. A spring with damping satisfies $y'' + 2y' + 5y = 5$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 5y = 5$$

Answer: _____



6. Find the steady-state (particular) solution of $y'' + 3y' + 3y = 16$.

$$y'' + 3y' + 3y = 16$$

Answer: _____

7. Find a particular solution y_p for $y'' + 5y' + 3y = 6$.

$$y'' + 5y' + 3y = 6$$

Answer: _____

8. A spring with damping satisfies $y'' + 2y' + 4y = 17$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 17$$

Answer: _____

9. Find the steady-state (particular) solution of $y'' + 4y' + 9y = 11$.

$$y'' + 4y' + 9y = 11$$

Answer: _____

10. Find a particular solution y_p for $y'' + 2y' + 5y = 19$.

$$y'' + 2y' + 5y = 19$$

Answer: _____

11. A spring with damping satisfies $y'' + 4y' + 2y = 20$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 4y' + 2y = 20$$

Answer: _____

12. Find the steady-state (particular) solution of $y'' + 5y' + 9y = 9$.

$$y'' + 5y' + 9y = 9$$

Answer: _____

13. Find a particular solution y_p for $y'' + 2y' + 4y = 19$.

$$y'' + 2y' + 4y = 19$$

Answer: _____



14. A spring with damping satisfies $y'' + 2y' + 3y = 8$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 3y = 8$$

Answer: _____

15. Find the steady-state (particular) solution of $y'' + 5y' + 3y = 18$.

$$y'' + 5y' + 3y = 18$$

Answer: _____

16. Find a particular solution y_p for $y'' + 1y' + 3y = 20$.

$$y'' + 1y' + 3y = 20$$

Answer: _____

17. A spring with damping satisfies $y'' + 6y' + 3y = 10$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 6y' + 3y = 10$$

Answer: _____

18. Find the steady-state (particular) solution of $y'' + 2y' + 3y = 23$.

$$y'' + 2y' + 3y = 23$$

Answer: _____

19. Find a particular solution y_p for $y'' + 3y' + 3y = 20$.

$$y'' + 3y' + 3y = 20$$

Answer: _____

20. A spring with damping satisfies $y'' + 5y' + 3y = 21$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 5y' + 3y = 21$$

Answer: _____

21. Find the steady-state (particular) solution of $y'' + 5y' + 4y = 5$.

$$y'' + 5y' + 4y = 5$$

Answer: _____



22. Find a particular solution y_p for $y'' + 4y' + 4y = 11$.

$$y'' + 4y' + 4y = 11$$

Answer: _____

23. A spring with damping satisfies $y'' + 5y' + 2y = 24$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 5y' + 2y = 24$$

Answer: _____

24. Find the steady-state (particular) solution of $y'' + 4y' + 5y = 12$.

$$y'' + 4y' + 5y = 12$$

Answer: _____

25. Find a particular solution y_p for $y'' + 1y' + 3y = 15$.

$$y'' + 1y' + 3y = 15$$

Answer: _____

26. A spring with damping satisfies $y'' + 2y' + 2y = 17$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 2y = 17$$

Answer: _____

27. Find the steady-state (particular) solution of $y'' + 4y' + 8y = 26$.

$$y'' + 4y' + 8y = 26$$

Answer: _____

28. Find a particular solution y_p for $y'' + 6y' + 1y = 5$.

$$y'' + 6y' + 1y = 5$$

Answer: _____

29. A spring with damping satisfies $y'' + 2y' + 4y = 18$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 18$$

Answer: _____



30. Find the steady-state (particular) solution of $y'' + 3y' + 7y = 22$.

$$y'' + 3y' + 7y = 22$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Particular solution — constant forcing. All answers verified by independent computation.

Solutions



Particular solution — constant forcing

1. Find a particular solution y_p for $y'' + 6y' + 1y = 2$.

$$y'' + 6y' + 1y = 2$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 6 \cdot 0 + 1 \cdot A = 2$.

→ Solving: $A = 2/1 = 2$.

→ Particular solution: $y_p = 2$.

Answer: $y_p = 2$

2. A spring with damping satisfies $y'' + 2y' + 4y = 8$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 8$$

→ At equilibrium, $y'' = y' = 0$. So $4 \cdot y_p = 8$.

→ $y_p = 8/4 = 2$.

Answer: $y_p = 2$

3. Find the steady-state (particular) solution of $y'' + 4y' + 7y = 20$.

$$y'' + 4y' + 7y = 20$$

→ For constant forcing $f = 20$, try $y_p = A$.

→ Substituting: $7A = 20$, so $A = 20/7 = 20/7$.

→ Steady-state solution: $y_p = 20/7$.

Answer: $y_p = \frac{20}{7}$

4. Find a particular solution y_p for $y'' + 3y' + 4y = 17$.

$$y'' + 3y' + 4y = 17$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 3 \cdot 0 + 4 \cdot A = 17$.

→ Solving: $A = 17/4 = 17/4$.

→ Particular solution: $y_p = 17/4$.

Answer: $y_p = \frac{17}{4}$

5. A spring with damping satisfies $y'' + 2y' + 5y = 5$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 5y = 5$$

→ At equilibrium, $y'' = y' = 0$. So $5 \cdot y_p = 5$.

→ $y_p = 5/5 = 1$.

Answer: $y_p = 1$



6. Find the steady-state (particular) solution of $y'' + 3y' + 3y = 16$.

$$y'' + 3y' + 3y = 16$$

→ For constant forcing $f = 16$, try $y_p = A$.

→ Substituting: $3A = 16$, so $A = 16/3 = 16/3$.

→ Steady-state solution: $y_p = 16/3$.

Answer: $y_p = \frac{16}{3}$

7. Find a particular solution y_p for $y'' + 5y' + 3y = 6$.

$$y'' + 5y' + 3y = 6$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 5 \cdot 0 + 3 \cdot A = 6$.

→ Solving: $A = 6/3 = 2$.

→ Particular solution: $y_p = 2$.

Answer: $y_p = 2$

8. A spring with damping satisfies $y'' + 2y' + 4y = 17$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 17$$

→ At equilibrium, $y'' = y' = 0$. So $4 \cdot y_p = 17$.

→ $y_p = 17/4 = 17/4$.

Answer: $y_p = \frac{17}{4}$

9. Find the steady-state (particular) solution of $y'' + 4y' + 9y = 11$.

$$y'' + 4y' + 9y = 11$$

→ For constant forcing $f = 11$, try $y_p = A$.

→ Substituting: $9A = 11$, so $A = 11/9 = 11/9$.

→ Steady-state solution: $y_p = 11/9$.

Answer: $y_p = \frac{11}{9}$

10. Find a particular solution y_p for $y'' + 2y' + 5y = 19$.

$$y'' + 2y' + 5y = 19$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 2 \cdot 0 + 5 \cdot A = 19$.

→ Solving: $A = 19/5 = 19/5$.

→ Particular solution: $y_p = 19/5$.

Answer: $y_p = \frac{19}{5}$

11. A spring with damping satisfies $y'' + 4y' + 2y = 20$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 4y' + 2y = 20$$

→ At equilibrium, $y'' = y' = 0$. So $2 \cdot y_p = 20$.

→ $y_p = 20/2 = 10$.

Answer: $y_p = 10$



12. Find the steady-state (particular) solution of $y'' + 5y' + 9y = 9$.

$$y'' + 5y' + 9y = 9$$

→ For constant forcing $f = 9$, try $y_p = A$.

→ Substituting: $9A = 9$, so $A = 9/9 = 1$.

→ Steady-state solution: $y_p = 1$.

Answer: $y_p = 1$

13. Find a particular solution y_p for $y'' + 2y' + 4y = 19$.

$$y'' + 2y' + 4y = 19$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 2 \cdot 0 + 4 \cdot A = 19$.

→ Solving: $A = 19/4 = 19/4$.

→ Particular solution: $y_p = 19/4$.

Answer: $y_p = \frac{19}{4}$

14. A spring with damping satisfies $y'' + 2y' + 3y = 8$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 3y = 8$$

→ At equilibrium, $y'' = y' = 0$. So $3 \cdot y_p = 8$.

→ $y_p = 8/3 = 8/3$.

Answer: $y_p = \frac{8}{3}$

15. Find the steady-state (particular) solution of $y'' + 5y' + 3y = 18$.

$$y'' + 5y' + 3y = 18$$

→ For constant forcing $f = 18$, try $y_p = A$.

→ Substituting: $3A = 18$, so $A = 18/3 = 6$.

→ Steady-state solution: $y_p = 6$.

Answer: $y_p = 6$

16. Find a particular solution y_p for $y'' + 1y' + 3y = 20$.

$$y'' + 1y' + 3y = 20$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 1 \cdot 0 + 3 \cdot A = 20$.

→ Solving: $A = 20/3 = 20/3$.

→ Particular solution: $y_p = 20/3$.

Answer: $y_p = \frac{20}{3}$

17. A spring with damping satisfies $y'' + 6y' + 3y = 10$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 6y' + 3y = 10$$

→ At equilibrium, $y'' = y' = 0$. So $3 \cdot y_p = 10$.

→ $y_p = 10/3 = 10/3$.

Answer: $y_p = \frac{10}{3}$



18. Find the steady-state (particular) solution of $y'' + 2y' + 3y = 23$.

$$y'' + 2y' + 3y = 23$$

→ For constant forcing $f = 23$, try $y_p = A$.

→ Substituting: $3A = 23$, so $A = 23/3 = 23/3$.

→ Steady-state solution: $y_p = 23/3$.

Answer: $y_p = \frac{23}{3}$

19. Find a particular solution y_p for $y'' + 3y' + 3y = 20$.

$$y'' + 3y' + 3y = 20$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 3 \cdot 0 + 3 \cdot A = 20$.

→ Solving: $A = 20/3 = 20/3$.

→ Particular solution: $y_p = 20/3$.

Answer: $y_p = \frac{20}{3}$

20. A spring with damping satisfies $y'' + 5y' + 3y = 21$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 5y' + 3y = 21$$

→ At equilibrium, $y'' = y' = 0$. So $3 \cdot y_p = 21$.

→ $y_p = 21/3 = 7$.

Answer: $y_p = 7$

21. Find the steady-state (particular) solution of $y'' + 5y' + 4y = 5$.

$$y'' + 5y' + 4y = 5$$

→ For constant forcing $f = 5$, try $y_p = A$.

→ Substituting: $4A = 5$, so $A = 5/4 = 5/4$.

→ Steady-state solution: $y_p = 5/4$.

Answer: $y_p = \frac{5}{4}$

22. Find a particular solution y_p for $y'' + 4y' + 4y = 11$.

$$y'' + 4y' + 4y = 11$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 4 \cdot 0 + 4 \cdot A = 11$.

→ Solving: $A = 11/4 = 11/4$.

→ Particular solution: $y_p = 11/4$.

Answer: $y_p = \frac{11}{4}$

23. A spring with damping satisfies $y'' + 5y' + 2y = 24$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 5y' + 2y = 24$$

→ At equilibrium, $y'' = y' = 0$. So $2 \cdot y_p = 24$.

→ $y_p = 24/2 = 12$.

Answer: $y_p = 12$



24. Find the steady-state (particular) solution of $y'' + 4y' + 5y = 12$.

$$y'' + 4y' + 5y = 12$$

→ For constant forcing $f = 12$, try $y_p = A$.

→ Substituting: $5A = 12$, so $A = 12/5 = 12/5$.

→ Steady-state solution: $y_p = 12/5$.

Answer: $y_p = \frac{12}{5}$

25. Find a particular solution y_p for $y'' + 1y' + 3y = 15$.

$$y'' + 1y' + 3y = 15$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 1 \cdot 0 + 3 \cdot A = 15$.

→ Solving: $A = 15/3 = 5$.

→ Particular solution: $y_p = 5$.

Answer: $y_p = 5$

26. A spring with damping satisfies $y'' + 2y' + 2y = 17$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 2y = 17$$

→ At equilibrium, $y'' = y' = 0$. So $2 \cdot y_p = 17$.

→ $y_p = 17/2 = 17/2$.

Answer: $y_p = \frac{17}{2}$

27. Find the steady-state (particular) solution of $y'' + 4y' + 8y = 26$.

$$y'' + 4y' + 8y = 26$$

→ For constant forcing $f = 26$, try $y_p = A$.

→ Substituting: $8A = 26$, so $A = 26/8 = 13/4$.

→ Steady-state solution: $y_p = 13/4$.

Answer: $y_p = \frac{13}{4}$

28. Find a particular solution y_p for $y'' + 6y' + 1y = 5$.

$$y'' + 6y' + 1y = 5$$

→ Try $y_p = A$ (constant). Substituting into the ODE: $0 + 6 \cdot 0 + 1 \cdot A = 5$.

→ Solving: $A = 5/1 = 5$.

→ Particular solution: $y_p = 5$.

Answer: $y_p = 5$

29. A spring with damping satisfies $y'' + 2y' + 4y = 18$ (constant forcing). Find the equilibrium (particular) solution.

$$y'' + 2y' + 4y = 18$$

→ At equilibrium, $y'' = y' = 0$. So $4 \cdot y_p = 18$.

→ $y_p = 18/4 = 9/2$.

Answer: $y_p = \frac{9}{2}$



30. Find the steady-state (particular) solution of $y'' + 3y' + 7y = 22$.

$$y'' + 3y' + 7y = 22$$

→ For constant forcing $f = 22$, try $y_p = A$.

→ Substituting: $7A = 22$, so $A = 22/7 = 22/7$.

→ Steady-state solution: $y_p = 22/7$.

Answer: $y_p = \frac{22}{7}$

